

Battery TS + EIS + DRT Research

Prepared by Nayeem Faisal | 24 September 2026 | Evidence-led source intake and next research decision

Current position The project has a public research record, source evidence ledger, DRT evaluation protocol, and local structural EIS intake prototype.	New research increment Two supervisor-recommended TUM ageing sources have been screened and assigned specific roles without overstating their EIS coverage.
What is ready Controlled file-level audit and adapter work for one direct TS + EIS candidate.	Scientific boundary No source is called a TS + EIS + DRT match until file-level identifiers, state, protocol, units, and impedance coverage are verified.

WORK COMPLETED

From literature to a testable research workflow

- Reviewed the supervisor-provided battery-data paper and converted it into four quality gates: source provenance, experiment context, a proven TS-to-EIS join, and DRT quality.
- Added the TUM LG MJ1 ageing, charging, BEV-profile, and OCV source to the candidate registry and evidence ledger.
- Added the TUM 196-cell ageing campaign with repeated SOH and degradation-mode labels to the candidate registry and evidence ledger.
- Updated the dated public wiki pages so all conclusions and pending checks remain reviewable.

How the new TUM sources support the project

Source	Verified contribution	Role in this project	Boundary / next audit
LG MJ1 ageing, charging, and BEV profiles	10 LG Chem INR18650 MJ1 cells; capacity/resistance ageing, BEV profiles, charging curves, quasi-stationary OCV.	Time-series ingestion, ageing-state metadata, partial-charge and OCV features.	No EIS pairing established. Audit README, identifiers, and archive layout.
196-cell experimental degradation study	Calendar, cyclic, and dynamic ageing; periodic capacity, resistance, LLI, LAM_NE, and LAM_PE labels.	Large TS-to-SOH and degradation-label reference; protocol-sensitivity benchmark.	No EIS pairing established. Audit time-series/check-up keys and look for explicit impedance material.
Direct TS + EIS queue	LFP EIS/pulse, fast-charge ageing, 48-cell degradation path, online EIS prediction, and A123 multimodal candidates.	Controlled source-level TS-to-EIS matching and later DRT evaluation.	Start with one small lawful sample and a file manifest.

NEXT CONTROLLED TASK

One source, one manifest, one evidence decision

- Select one direct candidate that has both time-series and EIS evidence.
- Download the smallest lawful sample and run the read-only source inventory.
- Record cell, test, cycle, state, temperature, protocol, units, licence, and checksum fields.
- Decide pass, hold, or reject for the TS-to-EIS join. Preserve negative findings.
- Only a supported join proceeds to structural EIS intake and DRT evaluation.

Decision requested: confirm whether the first experiment should prioritise state-aware matching, ageing-labelled modelling, or an EIS/DRT method benchmark.

Public wiki: <https://nayeemfaisal.github.io/LLM-Wiki-Site/>

Key sources: <https://doi.org/10.3390/batteries9070385> | <https://doi.org/10.1016/j.est.2022.106517> | <https://doi.org/10.1016/j.jpowsour.2022.232498>